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Leakage-Resilient k -out-of- n Oblivious Transfer*

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Abstract—Oblivious transfer (OT) is an important cryptographic primitive. In the traditional OT protocol, it requires that an adversary $\mathcal{A}$ can see the messages sent and received by the uncorrupted party, but the party's internal state is completely confidential.

In this paper, we consider OT protocols in the presence of information leakage. Informally, the adversary $\mathcal{A}$ is allowed to obtain arbitrary bounded leakage on the uncorrupted party's internal state. We first propose a formal definition of leakage-resilient k -out-of- n oblivious transfer. Subsequently, inspired by the idea from Green et al. (Asiacrypt '07), where they construct OT protocols based on identify-based encryption (IBE), we present the first non-adaptive and adaptive leakage-resilient k -out-of- n OT protocols based on leakage-resilient IBE which meet the proposed definition.

Index Terms—Adaptive oblivious transfer, Identify-based encryption, Leakage-resilient OT, Dual system encryption.

I. INTRODUCTION

Oblivious transfer (OT), introduced by Rabin [1], is an important cryptographic primitive. It is one of the basic and important building blocks for many cryptographic schemes, especially it is a key block for secure two-party or multi-party computation [2], [3]. Informally, in an OT protocol, there are two parties, the sender Sen who has some messages and the receiver Rec who wants to obtain some of the sender's messages. In a k -out-of- n oblivious transfer protocol (denoted by OT_k^n), Sen owns n messages $M_1, \dots, M_n$ and Rec owns k indexes $\sigma_1, \dots, \sigma_k \in [1, n]$. The aim of the protocol is for Rec to receive messages $M_{\sigma_1}, \dots, M_{\sigma_k}$ and learn nothing about M_i , ($i \in [1, n] \setminus \{\sigma_1, \dots, \sigma_k\}$). Meanwhile, Rec reveals nothing about indexes $\sigma_1, \dots, \sigma_k \in [1, n]$ to Sen during the execution. Adaptive oblivious transfer was proposed by Naor and Pinkas [4], where the receiver Rec may decide on σ_i based on $M_{\sigma_1}, \dots, M_{\sigma_{i-1}}$. There are several different adaptive OT_k^n protocols based on different cryptography tools [5].

In the traditional OT protocols, we assume that the uncorrupted parties' internal state is completely inaccessible by the adversary. Unfortunately, over the last two decades, researchers have begun to realize that the adversary might get some partial information about the internal state of the uncorrupted party via various side-channel attacks [6]. In this

work, we consider an adversary, in addition of corrupting a party, is able to observe bounded leakage on the internal state of the uncorrupted parties. Leakage is modeled by allowing adversary to adaptive submit leakage functions which is applied to the uncorrupted parties' internal state. In our paper, we follow the work of Dziembowski and Faust [7], and make the following two restrictions on the leakage:

- 1. Bounded leakage: the amount of leakage in each implementation is bounded in length;
- 2. Independent leakage: the computation can be structured into steps, where each such step leaks independently.

According to the second restriction, we divide the memory of a device into two parts, an active part and a passive part. The active part is the memory part which is currently accessed by a computation and adversary can obtain the leakage information only about the active part at any given time. Similar restrictions have been used in several previous works.

In the past few years, there has been an extensive amount of research to construct leakage-resilient cryptographic primitives, such as leakage-resilient encryption schemes [8], leakage-resilient signature schemes [9]. Most recently, the leakage-resilient interactive protocols has drawn increasing attention [10]. But, there has very limited research on leakage-resilient oblivious transfer (denoted by LR-OT). To the best of our knowledge, there are only four existing leakage-resilient 1-out-of-2 OT protocols [11]–[14]. In [13], Damgård et al. constructed a leakage-resilient 1-out-of-2 OT by combining the OT protocol presented by Peikert et al. [15] with the encryption scheme from [16]. But their protocol was applied in the only computational leakage model of [17], and the security was proved in the semi-honest model. In [11], Bitansky et al. presented a leakage-resilient 1-out-of-2 OT based on non-committing encryption [16]. Their protocol is secure against semi-honest adversary in the UC framework. In [12], Boyle et al. constructed an augmented non-committing encryption such that the public keys generated by the oblivious key generation algorithm with overwhelming probability were lossy. Meanwhile, they presented a special randomness generation procedure in the CRS model. Based on the two tools, they constructed a leakage-resilient 1-out-of-4 OT protocol which was proved to be secure against bounded leakage in the

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TABLE I
COMPARING THE FOUR PROTOCOLS

Protocols	Type	Security model	Adversary
DHP11	1-out-of-2	CRS	semi-honest
BCH12	1-out-of-2	CRS	semi-honest
BGJ+13	1-out-of-4	CRS	semi-malicious
RA14	1-out-of-2	Standard	malicious
LR- OT_k^n	k-out-of-n	Standard	malicious
Adaptive LR- OT_k^n	k-out-of-n	RO	malicious

CRS model. In their protocol, the adversary was assumed to be semi-malicious, it follows the protocol specification may adaptively choose its input and randomness for any protocol step following any arbitrary *PPT* strategy. In [14], Roy et al. formalize the definition and propose a construction of a one-sided leakage-resilient privacy only two-message oblivious transfer protocol. In this paper, we will construct a leakage-resilient k-out-of-n OT protocol and a leakage-resilient adaptive k-out-of-n OT protocol. We compare our protocols to the old ones shown in Table I.

Our contribution. In this work, we first formalize the definition of leakage-resilient oblivious transfer within the real/ideal model paradigm. Then, we present the first efficient LR- OT_k^n protocol and adaptive LR- OT_k^n protocol. A brief introduction of our results are as follows.

In [18], Green and Hohenberger provided non-adaptive and adaptive OT_k^n protocols based on the IBE system and under the static intractability assumption DBDH. In [19], Lewko et al. proposed a leakage-resilient IBE which is resilient to bounded leakage from each of many secret keys per user, as well as many master keys. Based on their results, we want to construct OT_k^n protocols which are resilient to bounded leakage. In our non-adaptive construction, *Sen* uses the leakage-resilient encryption algorithm to encrypt the messages and sends these ciphertexts to receiver *Rec*. Then, *Rec* adopts leakage-resilient blind extraction algorithm, blind extraction algorithm was defined by [18], to obtain the corresponding decryption keys. Here, *Rec* will employ the leakage-resilient identification protocol in [7] to prove the correctness of the messages he was sending during the extraction process. After *Rec* extracts the corresponding keys, he can obtain the corresponding messages by decrypting the corresponding ciphertexts. Finally, we prove that our protocol is secure in the bounded and independently leakage model, when there is a malicious receiver or a semi-honest sender. Since any one-side simulatable OT protocol can be transformed to full simulatable OT with the help of Zero-knowledge proof(ZKP), so our scheme can be transformed into a full-simulatable protocol as long as the ZKPs for the sender's messages are leakage-resilient ZKPs. To the best of our knowledge, this is the first construction of the LR- OT_k^n protocol. Then, we construct an adaptive LR- OT_k^n protocol by making a modification to the non-adaptive protocol. The adaptive protocol is secure in the random oracle model.

Organization. This article is organized as follows. First, we present some preliminaries that are used in our construction

in Section 2. In Section 3, we propose the formal definition of leakage-resilient oblivious transfer in the ideal model paradigm. Based on this, we present in Section 4 the concrete construction of our LR- OT_k^n protocol and the security proof of it. Furthermore, we construct an adaptive LR- OT_k^n protocol in Section 5. Finally, Section 6 is conclusion.

II. PRELIMINARY

Notations. We use $a \leftarrow_R A$ denote that a is picked uniformly at random from the set A , and $a, b, c \leftarrow_R A$ denote that a, b, c are picked uniformly at random from the set A . Let g be a group element and $\vec{\rho} = \langle \rho_1, \dots, \rho_n \rangle$ be a vector, we denote by $g^{\vec{\rho}}$ the vector $C = \langle g^{\rho_1}, \dots, g^{\rho_n} \rangle$. Furthermore, let $B = \langle B_1, \dots, B_n \rangle$ be a vector, we denote C^B as the vector $\langle g^{\rho_1 B_1}, \dots, g^{\rho_n B_n} \rangle$. We denote $*$ to be the component-wise multiplication of two vectors. Let $\mathbb{N}$ be the set of natural numbers, for $n \in \mathbb{N}$, we denote $[1, n]$ for $\{1, \dots, n\}$ and $\text{negl}(n)$ for a negligible function of n . We write *PPT* for probabilistic polynomial time. And, by $X \approx_c Y$, we denote the random variables X, Y are computational undistinguishable.

A. Composite Order Bilinear Groups and Some Computational Assumption

Let $\mathcal{G}$ be a group generator that, on input the security parameter 1^λ where $\lambda \in \mathbb{N}$, outputs the description of a bilinear mapping as $(N = p_1 p_2 p_3, \mathbb{G}, \mathbb{G}_T, e(\cdot, \cdot))$, where p_1, p_2, p_3 are distinct prime numbers of magnitude $\Theta(2^\lambda)$, both $\mathbb{G}$ and $\mathbb{G}_T$ are cyclic groups of order N , and $e : \mathbb{G} \times \mathbb{G} \mapsto \mathbb{G}_T$ is a bilinear map such that $\forall g \in \mathbb{G}$ and $a, b \in \mathbb{Z}_N$, $e(g^a, g^b) = e(g, g)^{ab}$. We denote by $\mathbb{G}_i$ the subgroup of $\mathbb{G}$ of order p_i , ($i = 1, 2, 3$). Obviously, for every $g_i \in \mathbb{G}_i, g_j \in \mathbb{G}_j$, if $i \neq j$, we have $e(g_i, g_j) = 1$. We use $\mathbb{G}_{12}$ to denote the subgroup of $\mathbb{G}$ of order $p_1 p_2$. We also define $\mathbb{G}_{13}, \mathbb{G}_{23}$ similarly. For vectors $\vec{a} = \langle a_1, \dots, a_n \rangle \in \mathbb{G}^n$ and $\vec{b} = \langle b_1, \dots, b_n \rangle \in \mathbb{G}^n$, we define $e_n(\vec{a}, \vec{b}) := \prod_{i=1}^n e(a_i, b_i) \in \mathbb{G}_T$ [19].

We define the variables $PP = (N, \mathbb{G}, \mathbb{G}_T, e(\cdot, \cdot)) \leftarrow \mathcal{G}(1^\lambda)$ and $X_1 \leftarrow_R \mathbb{G}_1, X_2 \leftarrow_R \mathbb{G}_2, X_3 \leftarrow_R \mathbb{G}_3$. Lewko et al. [19] showed the following assumptions:

Assumption 1. Given PP and $D^{(1)} = (X_1, X_3)$, no *PPT* adversary has a non-negligible advantage in distinguishing $T_0^{(1)} \leftarrow_R \mathbb{G}_{12}$ from $T_1^{(1)} \leftarrow_R \mathbb{G}_1$. We define the advantage of an adversary $\mathcal{A}$ in solving Assumption 1 as:

$$\text{Adv}_{\mathcal{A}}^{(1)}(\lambda) = \left| \frac{\Pr[\mathcal{A}(PP, D^{(1)}, T_0^{(1)}) = 1]}{\Pr[\mathcal{A}(PP, D^{(1)}, T_1^{(1)}) = 1]} - 1 \right|.$$

Assumption 2. The advantage of any *PPT* adversary in distinguishing $T_0^{(2)} \leftarrow_R \mathbb{G}$ from $T_1^{(2)} \leftarrow_R \mathbb{G}_{13}$ is negligible, given PP and $D^{(2)} = (X_1, X_3, X_1^\alpha X_2^\beta, X_2^\gamma X_3^\delta)$ where $\alpha, \beta, \gamma, \delta \leftarrow_R \mathbb{Z}_N$. We define the advantage of an adversary $\mathcal{A}$ in solving Assumption 2 as:

$$\text{Adv}_{\mathcal{A}}^{(2)}(\lambda) = \left| \frac{\Pr[\mathcal{A}(PP, D^{(2)}, T_0^{(2)}) = 1]}{\Pr[\mathcal{A}(PP, D^{(2)}, T_1^{(2)}) = 1]} - 1 \right|.$$

Assumption 3. The advantage for any *PPT* adversary to distinguish $T_0^{(3)} = e(X_1, X_1)^{\alpha\gamma}$ from $T_1^{(3)} \leftarrow_R \mathbb{G}_{13}$ is negligible, given PP and $D^{(3)} = (X_1, X_2, X_3, X_1^\alpha X_2^\beta, X_1^\gamma X_2^\delta)$

where $\alpha, \beta, \gamma, \delta \leftarrow_R \mathbb{Z}_N$. We define the advantage of an adversary $\mathcal{A}$ in solving Assumption 3 as:

$$\text{Adv}_{\mathcal{A}}^{(3)}(\lambda) = \left| \frac{\Pr[\mathcal{A}(PP, D^{(3)}, T_0^{(3)}) = 1]}{\Pr[\mathcal{A}(PP, D^{(3)}, T_1^{(3)}) = 1]} \right|.$$

We say Assumption i , ($i = 1, 2, 3$) holds if $\text{Adv}_{\mathcal{A}}^{(i)}(\lambda) \leq \text{negl}(\lambda)$ is satisfied for all PPT adversary $\mathcal{A}$.

B. Identity-Based Encryption

An IBE scheme, introduced by Shamir [2], consists of four PPT algorithms *Setup*, *Extract*, *Encrypt* and *Decrypt*.

- *Setup*(1^λ): On input a security parameter 1^λ , it produces a master public key mpk and a master secret key msk . Where the mpk defines the message space $\mathcal{M}$ and the identity set $\mathcal{ID}$.
- *Extract*(id, msk): On input an identity $id \in \mathcal{ID}$ and the msk , it generates an identity secret key sk_{id} .
- *Encrypt*(M, id): On input a message $M \in \mathcal{M}$ and an identity id , it outputs a ciphertext C of the message M for identity id .
- *Decrypt*(C, sk_{id}): On input a ciphertext C and an identity secret key sk_{id} , it outputs a message M' or $\perp$ indicating decryption failure.

We have $M = \text{Decrypt}(sk_{id}, \text{Encrypt}(M, id))$, for any $M \in \mathcal{M}$ and $id \in \mathcal{ID}$.

There are many different leakage-resilient IBE schemes [8], [19] according to different leakage models. But most of them only allow leakage on identity secret keys, but allow no leakage on the master secret key. In [19] and [20], the authors proposed IBE schemes which tolerate leakage on the master secret key separately. In our construction, we will adopt the technique in [19] to resist leakage.

C. Leakage-Resilient Okamoto Identification protocol

In our construction, *Rec* will adopt deformed Okamoto identification scheme [7] to prove the correctness of the messages he was sending during the extraction process. Let g_1 and g_2 be two generators of $\mathbb{G}$ and the discrete logarithm $\log_{g_1} g_2$ is unknown. Then, the prover $\mathcal{P}$ has the secret key $sk = (x_1, x_2) \leftarrow \mathbb{Z}_p^2$, where prime p is the order of $\mathbb{G}$, and the public key $pk = g_1^{x_1} g_2^{x_2}$. The standard Okamoto identification scheme as follows:

1. $\mathcal{P}$ chooses $(w_1, w_2) \leftarrow \mathbb{Z}_p^2$, computes $a := g_1^{w_1} g_2^{w_2}$ and sends a to $\mathcal{V}$,
2. $\mathcal{V}$ chooses $c \leftarrow \mathbb{Z}_p$ and sends it to $\mathcal{P}$,
3. $\mathcal{P}$ computes $z_1 = w_1 + cx_1$, $z_2 = w_2 + cx_2$ and sends (z_1, z_2) to $\mathcal{V}$,
4. $\mathcal{V}$ accepts if and only if $g_1^{z_1} g_2^{z_2} = a \cdot pk^c$.

In [7], Dziembowski et al. proposed a leakage-resilient Okamoto Identification protocol $\text{Oka}\{(x_1, x_2) | pk = g_1^{x_1} g_2^{x_2}\}$. In their construction, verification is exactly the same as in the standard Okamoto protocol, while the key generation and the computation of the prover is adjusted to protect against leakage attacks. Informally, instead of storing $(x_1, x_2) \in \mathbb{Z}_p^2$ in the memory, $\mathcal{P}$ selects a row vector $L \leftarrow \mathbb{Z}_p^n \setminus \{(0^n)\}$ and a

matrix $R \leftarrow \mathbb{Z}_p^{n \times 2}$ such that matrix product $L \times R = (x_1, x_2)$ and sets $sk = (L, R)$, $pk = g_1^{x_1} g_2^{x_2}$. We require that L and R leak independently. The concrete structure of the protocol $\text{Oka}\{(x_1, x_2) | pk = g_1^{x_1} g_2^{x_2}\}$ is given in Fig. 1.

In the bounded and independently leakage model, Dziembowski et al. proved that their protocol $\text{Oka}\{(x_1, x_2) | pk = g_1^{x_1} g_2^{x_2}\}$ can tolerate $\lambda = (0.15n - 3) \log p - 1$ bits leakage on the prover's secret key under the DL assumption.

III. LEAKAGE-RESILIENT K-OUT-OF-N OBLIVIOUS TRANSFER

In the traditional definition of security for OT via the real/ideal framework, we first consider the ideal scenario in which the sender and receiver send their inputs to an uncorrupted trusted third party. Then, the trusted party sends each party its respective output. Loosely speaking, an OT protocol is secure if, for any PPT adversary interacting in the real world we can construct an ideal world counterpart (called simulator) whose view is computationally indistinguishable from the view of the real adversary.

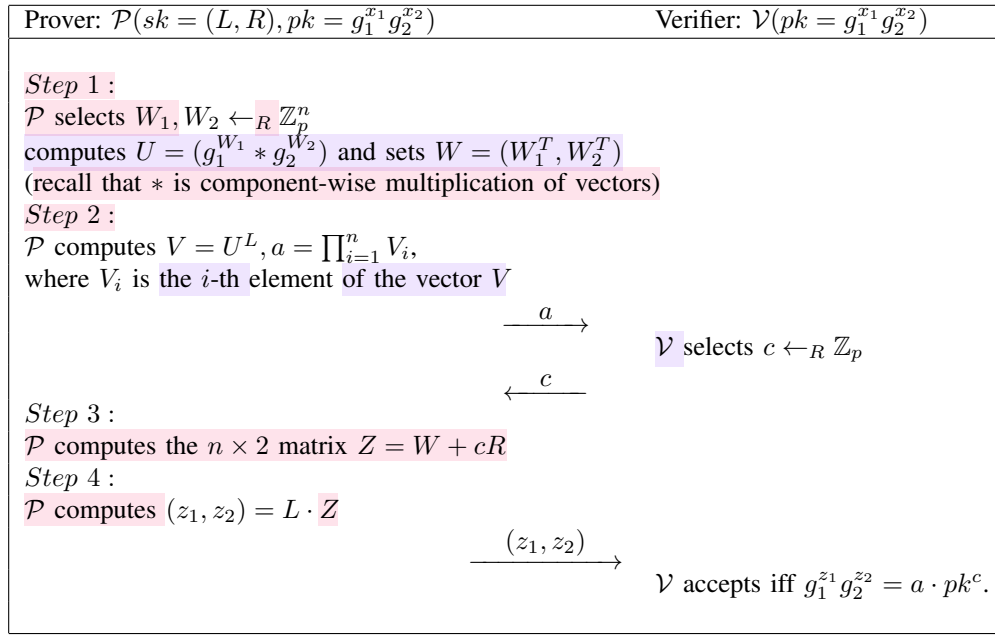
A. Leakage Model

We consider the environment where an adversary is allowed to obtain leakage from the entire state of the uncorrupted party, in addition to corrupting the other party. Where the party's entire state consists of his inputs and randomness. In this work, we split the memory of a physical device into two parts, an active and a passive part, which leak independently. The part being currently accessed by the computation is the active part. In other word, we structure any computation that involves access to the internal state into a sequence of steps. In each step, adversary can obtain leakage information only about one part of the memory (the active part). Adversary $\mathcal{A}$ is allowed to submit a sequence leakage functions $\{Leak_i\}$ applied on $x_i \in \{L, R\}$, where $L, R \in \{0, 1\}^n$ are the two parts of uncorrupted party's internal state. Suppose that $Leak_i : \{0, 1\}^n \rightarrow \{0, 1\}^{\lambda_i}$, the only restriction is that $\sum_j \lambda_j(L) \leq \lambda$ and $\sum_k \lambda_k(R) \leq \lambda$, that is $\mathcal{A}$ can not obtain more than λ bits information from each L and R . We refer to such $\mathcal{A}$ as a λ -limited adversary. For the sake of simplicity, we can assume that the adversary submits a single query $Leak(\cdot)$ since any arbitrary polynomial number of leakage queries can be simply hardwired into a single query. Therefore, we can denote as $Leak(L, R) = \{L_1(x_1), L_2(x_2), \dots\}$.

B. Definition.

For consistency with the earlier work, we use the notation from Camenish et al. [21] and Green et al. [18].

Definition 3.1(k-out-of-n Oblivious Transfer). An oblivious transfer protocol is a tuple of four algorithms (S_I, R_I, S_T, R_T) . During the initialization phase, the sender and receiver perform an interactive protocol where the sender *Sen* runs the S_I algorithm on input messages $M_1, \dots, M_n$ to obtain state value S_0 and the receiver *Rec* runs R_I algorithm without input to obtain state value R_0 . Next, during the transfer

Fig. 1. Protocol $Oka\{(x_1, x_2) | pk = g_1^{x_1} g_2^{x_2}\}$

phase, Sen and Rec interactively execute S_T, R_T respectively, details as follows.

Adaptive OT_k^n : During the i -th transfer, $1 \leq i \leq k$, Sen runs $S_T(S_{i-1})$ algorithm to obtain state value S_i , while Rec runs $R_T(R_{i-1}, \sigma_i)$, where $\sigma_i \in [1, n]$ is the index of the message to be received, to obtain state information R_i and the retrieved message M'_{σ_i} or $\perp$ indicating failure.

Non-adaptive OT_k^n : In this case, Sen and Rec execute the protocol as in the previous case except that the algorithm $R_T(R_{i-1}, \sigma_i)$ does not output a message for $1 \leq i < k$. Instead of this, $R_T(R_{k-1}, \sigma_k)$ algorithm outputs the messages $M'_{\sigma_1}, \dots, M'_{\sigma_k}$, where $M'_{\sigma_j}, (1 \leq j \leq k)$ is a valid message or $\perp$.

Definition 3.2(Leakage-Resilient k-out-of-n Oblivious Transfer LR- OT_k^n). Security for adaptive and non-adaptive LR- OT_k^n is defined according to the real/ideal paradigm. Here, we describe a real experiment in which the parties run the protocol themselves, while in the ideal experiment the functionality is implemented through a trusted third party.

Real experiment. $Real_{Sen, Rec}(n, k, M_1, \dots, M_n, \Sigma)$ proceeds as follows. The possibly corrupted sender $\hat{Sen}$ is given messages $(M_1, \dots, M_n)$ as input and interacts with the possibly corrupted receiver $\hat{Rec}(\Sigma)$, where Σ is a selection algorithm that on input the full collection of messages thus far received, outputs the index σ_i of the next message to be queried. At the beginning of the experiment, Sen and Rec produce initial states S_0 and R_0 respectively. In the adaptive case, for $1 \leq i \leq k$, $\hat{Sen}$ computes $S_i \leftarrow \hat{Sen}(S_{i-1})$ and $\hat{Rec}$ computes $(R_i, M'_{\sigma_i}) \leftarrow \hat{Rec}(R_{i-1})$. In the non-adaptive case, $\hat{Rec}$ obtains no messages until the k -th round, and therefore the selection algorithm Σ must be non-adaptive. In order to incorporate leakage attacks, we allow the adversary $\mathcal{A}$ who corrupting one party to make adaptive leakage queries

on the state of the uncorrupted party during the protocol execution. As mentioned, we model these leakage attacks as a single query $Leak(\cdot)$. We define $Real_{Leak(\cdot)}^A(Sen, Rec)$ for the real execution when adversary $\mathcal{A}$ submits the leakage query $Leak(\cdot)$. It is defined as the outputs of the uncorrupted party and $\mathcal{A}$ in the real protocol.

Ideal experiment. In $Ideal_{\hat{Sen}', \hat{Rec}'}(n, k, M_1, \dots, M_n, \Sigma)$, the possibly corrupted sender $\hat{Sen}'$ generates messages $(M'_1, \dots, M'_n)$ and transmits them to a trusted party T . In the i -th round $\hat{Sen}'$ sends a bit b_i to T , the possibly corrupted receiver $\hat{Rec}'$ transmits σ'_i to T . In the adaptive case, if $b_i = 1$ and $\sigma'_i \in [1, n]$, then T sends $M_{\sigma'_i}$ to $\hat{Rec}'$; otherwise, T sends $\perp$ to the $\hat{Rec}'$. In the non-adaptive case, T does not give $\hat{Rec}'$ any response until the k -th round. In the ideal environment, we allow the simulator $\mathcal{S}$ to learn leakage information too. This can be formalized by allowing $\mathcal{S}$ to submit a leakage query $Leak^{sm}(\cdot)$ to be applied on the inputs of the uncorrupted party, since there is no randomness is used in the ideal environment. In order to ensure our security definition to be meaningful, we should place some restrictions on the simulator's leakage query. Because we have to ensure that the simulator does not obtain too much leakage. Therefore, we limit that $Leak(\cdot)$ and $Leak^{sm}(\cdot)$ have the same output length. Moreover, same as [22], we stress that the simulator only using the output of leakage query to answer the adversary's leakage query. Such simulator is called leakage-oblivious simulator in [22]. We denote $Ideal_{Leak^{sm}(\cdot)}^S(\hat{Sen}', \hat{Rec}')$ for the ideal execution when simulator $\mathcal{S}$ queries $Leak^{sm}(\cdot)$ to the trusted party. It is defined as the outputs of the uncorrupted party and $\mathcal{S}$ in the ideal environment.

Sender Security. We say that (adaptive or non-adaptive) OT_k^n is sender-secure if for any PPT adversary $\mathcal{A}$ who corrupts

receiver $\hat{Rec}$ and submits an efficiently computable leakage query $Leak(\cdot)$, there exists a PPT simulator $\mathcal{S}$ that submits efficiently computable leakage query $Leak^{sm}(\cdot)$ where $Leak(\cdot)$ and $Leak^{sm}(\cdot)$ have the same output length, such that

$$Real_{Leak(\cdot)}^A(\hat{Sen}, \hat{Rec}) \approx_c Ideal_{Leak^{sm}(\cdot)}^S(\hat{Sen}, \hat{Rec}')$$

for any PPT distinguisher.

Receiver Security. We say that (adaptive or non-adaptive) OT_k^n is receiver-secure if for any PPT adversary $\mathcal{A}$ who corrupts sender $\hat{Sen}$ and submits an efficiently computable leakage query $Leak(\cdot)$, there exists a PPT simulator $\mathcal{S}$ that submits efficiently computable leakage query $Leak^{sm}(\cdot)$ where $Leak(\cdot)$ and $Leak^{sm}(\cdot)$ have the same output length, such that

$$Real_{Leak(\cdot)}^A(\hat{Sen}, Rec) \approx_c Ideal_{Leak^{sm}(\cdot)}^S(\hat{Sen}', Rec)$$

for any PPT distinguisher.

(λ_s, λ_r) -leakage-resilient OT_k^n . An sender-secure and receiver-secure (adaptive or non-adaptive) OT_k^n protocol (Sen, Rec) is said (λ_s, λ_r) -leakage-resilient OT_k^n , if the adversary $\mathcal{A}$ is a λ_s -limited adversary when he corrupting the receiver $\hat{Rec}$ and he is a λ_r -limited adversary when corrupting $\hat{Sen}$.

IV. LEAKAGE-RESILIENT NON-ADAPTIVE OT_k^n

A. Construction

We now proceed to construct a leakage-resilient non-adaptive OT_k^n . One of the main building blocks in our construction is the LR-IBE system proposed by Lewko et al. [19] which secure under the three assumptions in composite order groups (see assumption 1,2,3). They use the methodology of dual system encryption to construct an IBE scheme which is resilient to bounded leakage from master secret key, as well as identity secret keys.

Another crucial building block in our construction is the BlindExtract protocol, introduced by Green et al. [18], where the user U with identity $id \in ID$ obtains the corresponding secret key sk_{id} and the master authority learns nothing about id . We will use the BlindExtract protocol in a leakage-resilient way. In our construction, for every $id = \sigma_i$, Rec only needs to prove that he knows (θ_{id}, id) and (θ'_{id}, id) for $a_i = g_1^{\theta_{id}} u_1^{id}$, $b_i = h_1^{\theta'_{id}} g_3^{id}$. These are clearly Okamoto identification protocols. Therefore, we adopt the above-mentioned leakage-resilient Okamoto identification protocol to prevent the leakage of Rec 's index σ_i . Upcoming articles will describe our construction in more detail.

In our protocol, let Sen and Rec respective denote sender and receiver, and Sen has n messages $M_1, \dots, M_n \in \mathbb{G}_T$ as well as Rec has k indexes $\sigma_1, \dots, \sigma_k \in [1, n]$.

- First of all, the sender Sen calls the *Setup* algorithm to obtain a master secret key msk and a master public key mpk and sends mpk to the receiver Rec .
- After receiving mpk , Rec extracts the corresponding identity secret keys for identities σ_i , $1 \leq i \leq k$ via

executing the leakage-resilient blind extraction protocol with Sen .

- Then, Sen encrypts all messages and sends all ciphertexts to Rec .
- Rec decrypts the corresponding ciphertexts to obtain messages $\{M_{\sigma_1}, \dots, M_{\sigma_k}\}$.

During the execution of the leakage-resilient blind extraction protocol, Rec must prove to Sen that the elements it was sending have the right form. Here, we adopt the leakage-resilient identification protocol *Oka* was proposed in [7]. The details of our OT_k^n protocol is given in Fig. 2.

B. Correctness and Security Analysis

Correctness. Note that $C_{id_0} = M_{id} \cdot (e(g_1, g_1)^\alpha)^{s_{id}}$, therefore equality $M_{id} = \frac{C_{id_0}}{e_{m+2}(\overrightarrow{A_{id}}, \overrightarrow{C_{id_1}})}$ holds up if and only if $e_{m+2}(\overrightarrow{A_{id}}, \overrightarrow{C_{id_1}}) = (e(g_1, g_1)^\alpha)^{s_{id}}$. According to the above construction:

$$\begin{aligned} \overrightarrow{A_{id}} &= \langle A_1, \dots, A_m, \frac{A_{m+1}}{A_{m+3}^{\theta_{id}} A_{m+4}^{\theta'_{id}}}, A_{m+2} \rangle \\ &= \langle g_1^{y_1+z_1} g_3^{\rho_1}, \dots, g_1^{y_m+z_m} g_3^{\rho_m}, B_1, B_2 \rangle \end{aligned}$$

where $B_1 = g_1^{\alpha - \sum_{i=1}^m x_i(y_i+z_i)} h_1^{-(r+r')} g_3^{\rho_{m+1}+\rho'_{m+1}} a_{id}^{-(r+r')} b_{id}^{-\rho_{m+3}}$, and $B_2 = g_1^{-\theta_{id}(r+r')} h_1^{-\theta'_{id}\rho_{m+3}}$. Moreover,

$$\overrightarrow{C_{id_1}} = \langle (g_1^{x_1})^{s_{id}}, \dots, (g_1^{x_m})^{s_{id}}, g_1^{s_{id}}, (u_1^{id} h_1)^{s_{id}} \rangle.$$

As a result,

$$\begin{aligned} e_{m+2}(\overrightarrow{A_{id}}, \overrightarrow{C_{id_1}}) &= e(g_1, g_1)^{s_{id} \cdot \sum_{i=1}^m x_i(y_i+z_i)} \cdot e(g_1, g_1)^{s_{id} \cdot (\alpha - \sum_{i=1}^m x_i(y_i+z_i))} \\ &\quad \cdot e(g_1, g_1)^{-s_{id} \cdot (r+r')} \cdot e(g_1, u_1)^{-s_{id} \cdot (r+r') \cdot id} \\ &\quad \cdot e(g_1, u_1)^{s_{id} \cdot (r+r') \cdot id} \\ &= e(g_1, g_1)^{\alpha \cdot s_{id}}. \end{aligned}$$

Here, we omit the parts of which include $e(g_1, g_3)$, $e(u_1, g_3)$ or $e(h_1, g_3)$, because they are equal to 1.

Security analysis. We prove the following theorem:

Theorem 4.1. If the assumption 1,2,3 and DL assumption hold, then LR- OT_k^n is one-side simulatable $(\log N - O(1), (0.15l - 3) \log N - 1)$ -leakage-resilient non-adaptive OT_k^n .

Proof: We separately prove security in the case where Rec is corrupted and the case where Sen is corrupted.

In the case of Rec is corrupted. Let $\mathcal{A}$ be a malicious adversary controlling $\hat{Rec}$. We construct a simulator $\mathcal{S}_{Rec}$ as follows.

- $\mathcal{S}_{Rec}$ invokes $\mathcal{A}$ upon its inputs. Then it calls *Setup* algorithm to get a master secret key msk and a master public key mpk . $\mathcal{S}_{Rec}$ sends mpk to $\mathcal{A}$.
- $\mathcal{S}_{Rec}$ receives (a_i, b_i) , $1 \leq i \leq k$ from $\mathcal{A}$ and executes *Oka* protocols [7] with $\mathcal{A}$. For every $i \in [1, k]$, denote the corresponding *Oka* protocol for a_i is Oka_{a_i} and for b_i is Oka_{b_i} . If one of them is incorrect, then $\mathcal{S}_{Rec}$ sends $\perp$ to $\mathcal{A}$ and continues the next *Oka* protocol. If both *Oka*

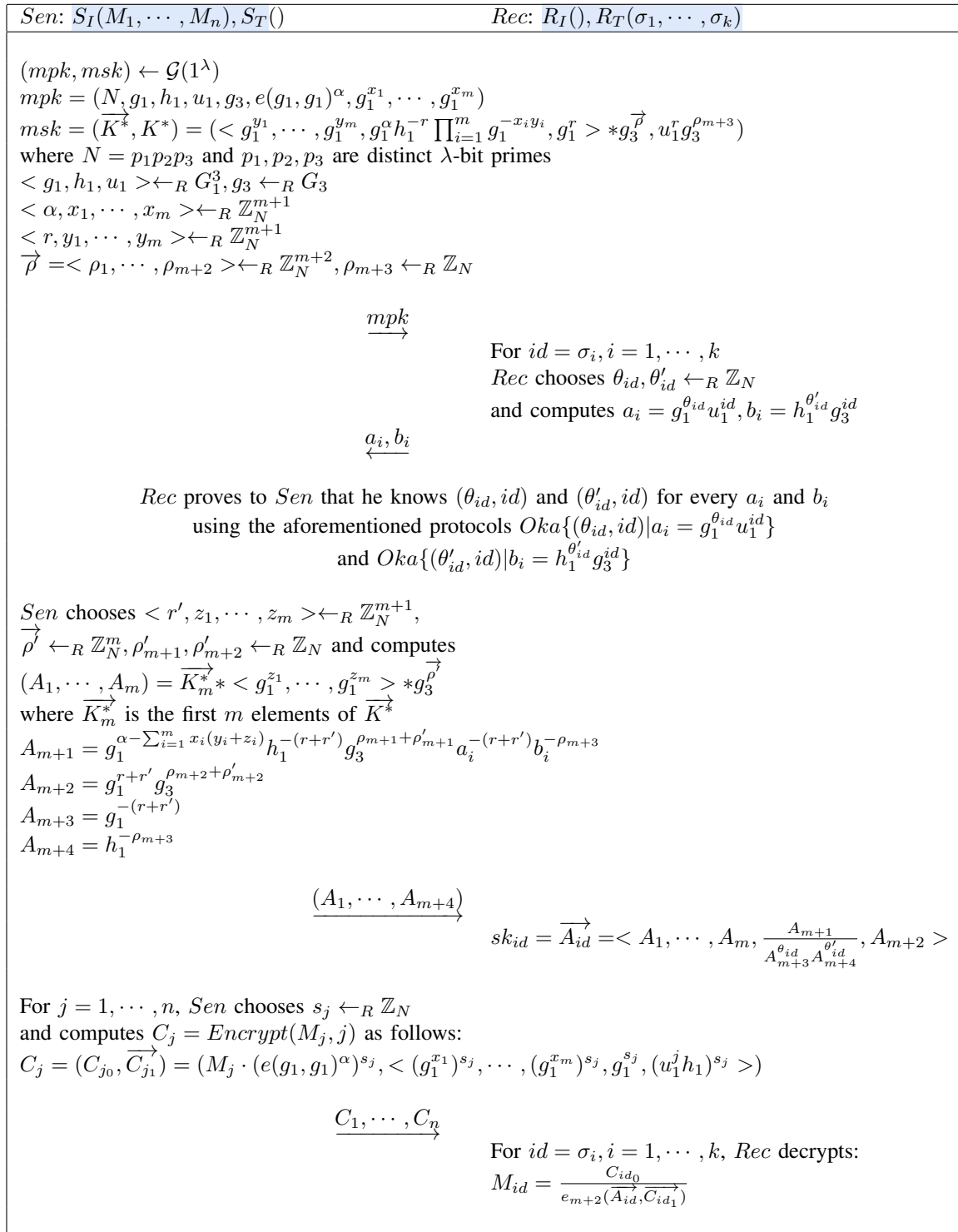


Fig. 2. LR-OT_k^m

protocols are correct for i , $\mathcal{S}_{Rec}$ can extract σ_i using the rewinding technology.

- After extracts $\sigma_1, \dots, \sigma_k$, $\mathcal{S}_{Rec}$ sends them to the trust third party and receives back the messages $M_{\sigma_1}, \dots, M_{\sigma_k}$.
- For $i \in \{\sigma_1, \dots, \sigma_k\}$, $\mathcal{S}_{Rec}$, as an honest *Sen* would,

computes $C_i = \text{Encrypt}(M_i, i) = (C_{i0}, \overrightarrow{C_{i1}})$. In contrast, it computes $C_i = \text{Encrypt}(0, i)$ for $i \in \mathcal{ID} \setminus \{\sigma_1, \dots, \sigma_k\}$. And then, $\mathcal{S}_{Rec}$ sends $C_1, \dots, C_n$ to $\mathcal{A}$.

- $\mathcal{S}_{Rec}$ outputs whatever $\mathcal{A}$ outputs and halts.

During the whole protocol, the sender's secret memory includes msk and messages $M_1, \dots, M_n$. So, if the adversary

$\mathcal{A}$ submits a leakage function $Leak(\cdot)$ at any time, $\mathcal{S}_{Rec}$ defines a new leakage function $Leak^{sm}(\cdot)$ so that $Leak^{sm}(\cdot)$ is incorporated with the same code of $Leak(\cdot)$ except that msk is already hardwired in it. Then $\mathcal{S}_{Rec}$ submits $Leak^{sm}(\cdot)$ to the trust party and receives back the leakage. Next, $\mathcal{S}_{Rec}$ hands $\mathcal{A}$ the leakage from the trust party. Because we employ the LR-IBE system in [19] where $l_{msk} = (m - 1 - 2c) \log P_2$, $c > 0$ is a fixed positive constant, we can take some suitable m such that $(m - 1 - 2c) \log P_2 > \log N - O(1)$, so the upper bound of the output length of $Leak(\cdot)$ is $\log N - O(1)$.

Recall that $Real_{Leak(\cdot)}^A(Sen, \hat{Rec})$ is the joint outputs of the parties in the real world and $Ideal_{Leak^{sm}(\cdot)}^{\mathcal{S}_{Rec}}(Sen, \hat{Rec})$ is the joint outputs of the parties in the ideal world. Then we continue with a proof that shows that:

$$Real_{Leak(\cdot)}^A(Sen, \hat{Rec}) \approx_c Ideal_{Leak^{sm}(\cdot)}^{\mathcal{S}_{Rec}}(Sen, \hat{Rec}).$$

That is, for any PPT distinguisher $\mathcal{D}$,

$$\left| \Pr[\mathcal{D}(Real_{Leak(\cdot)}^A(Sen, \hat{Rec})) = 1] - \Pr[\mathcal{D}(Ideal_{Leak^{sm}(\cdot)}^{\mathcal{S}_{Rec}}(Sen, \hat{Rec})) = 1] \right| \leq \text{negl}(\lambda).$$

To prove it, we begin with the real-world execution, and arrive at the ideal-world execution via a series of games. We denote $GAME_i$ as the joint outputs of the parties in Game i .

- Game 0: In this game, the honest sender Sen interacts with the corrupted receiver Rec . It's not hard to see, $GAME_0 = Real_{Leak(\cdot)}^A(Sen, \hat{Rec})$.
- Game 1: In this game, we use the rewinding technology to extract σ_i from the k sequential executions of the Oka protocols. If one of this extractions fails, we set Rec 's output to $\perp$. Therefore, $|\Pr[\mathcal{D}(GAME_1) = 1] - \Pr[\mathcal{D}(GAME_0) = 1]| \leq \text{negl}_1(\lambda)$.
- Game 2: In this game, we substitute the ciphertext vector $\langle C_1, \dots, C_n \rangle$ with a new vector $\langle C'_1, \dots, C'_n \rangle$, such that: for $i = 1, \dots, n$, if $i \in \{\sigma_1, \dots, \sigma_k\}$, we set $C'_i = C_i$, and otherwise set $C'_i \leftarrow \text{Encrypt}(0, i)$. According to [19], $|\Pr[\mathcal{D}(GAME_2) = 1] - \Pr[\mathcal{D}(GAME_1) = 1]| \leq \text{negl}_2(\lambda)$.

It is clear that $GAME_2 = Ideal_{Leak^{sm}(\cdot)}^{\mathcal{S}_{Rec}}(Sen, \hat{Rec})$, therefore we conclude that

$$\left| \Pr[\mathcal{D}(Real_{Leak(\cdot)}^A(Sen, \hat{Rec})) = 1] - \Pr[\mathcal{D}(Ideal_{Leak^{sm}(\cdot)}^{\mathcal{S}_{Rec}}(Sen, \hat{Rec})) = 1] \right| \leq \text{negl}_3(\lambda).$$

In the case of Sen is corrupted by a semi-honest adversary $\mathcal{A}$. Since Rec sends messages to Sen only in the k sequential Oka protocols, so we need only to call simulators in the Oka protocols to construct our simulator $\mathcal{S}_{Sen}$. Due to space constraints, we omit the description of the simulator $\mathcal{S}_{Sen}$, please refer to [7] for more details of $\mathcal{S}_{Sen}$.

In our construction, the receiver's secret memory consist of indexes $\sigma_1, \dots, \sigma_k$ and random values $y_i, y'_i, i \in [1, k]$. Therefore, when adversary $\mathcal{A}$ submits a leakage function $Leak(\cdot)$ at any time, $\mathcal{S}_{Sen}$ constructs a new leakage function $Leak^{sm}(\cdot)$ so that $Leak^{sm}(\cdot)$ possesses the same code of the $Leak(\cdot)$ except that $\{(y_i, y'_i) | i \in [1, k]\}$, used in simulation, are already hardwired in it. And then $\mathcal{S}_{Sen}$ submits $Leak^{sm}(\cdot)$

to the trust party and sends the leakage from the trust party to $\mathcal{A}$. Because we adopt the Oka protocol in [7] where the leakage bound is $(0.15l - 3) \log N - 1$, $l \in \mathbb{N}$, so the upper bound of the output length of $Leak(\cdot)$ is $(0.15l - 3) \log N - 1$.

According to the security of the Oka protocols in [7], we have that

$$Real_{Leak(\cdot)}^A(\hat{Sen}, Rec) \approx_c Ideal_{Leak^{sm}(\cdot)}^{\mathcal{S}_{Sen}}(\hat{Sen}, Rec),$$

and Theorem 4.1 follows. ■

V. LEAKAGE-RESILIENT ADAPTIVE OT_k^n

A. Construction

Intuitively, we can construct an adaptive leakage-resilient OT_k^n protocol by executing k times of the non-adaptive OT_1^n protocol. This can be realized in the standard model, but is inefficient for large k . So, same as [18], we construct an efficient adaptive leakage-resilient OT_k^n protocol in the random oracle model. In this construction, same as the previous protocol, we assume that Sen and Rec respective signify sender and receiver, and Sen has n messages $M_1, \dots, M_n \in \mathbb{G}_T$ as well as Rec has k indexes $\sigma_1, \dots, \sigma_k \in [1, n]$.

- Sen first selects $W_1, \dots, W_n \leftarrow_R \mathbb{G}_T$ and encrypts them with the encryption scheme which is used in the $LR-OT_k^n$ protocol. Let A_i be the ciphertext corresponding to W_i for any $i \in [1, n]$.
- Then, Sen computes $B_i = H(W_i) \oplus M_i$ for $1 \leq i \leq n$ and sets $C_i = (A_i, B_i)$, where $H : \mathbb{G}_T \rightarrow \mathbb{G}_T$ is a collision-resistant hash function. After that, Sen sends the public key mpk and the ciphertexts $C_1, \dots, C_n$ to Rec .
- After Rec received the public key and ciphertexts, the two parties run the leakage-resilient blind extraction which is used in our $LR-OT_k^n$ protocol for Rec to obtain the corresponding sk_{σ_i} .
- Then, Rec can get the corresponding message M_{σ_i} .

The details of our construction is given in Fig. 3, where $H : \mathbb{G}_T \rightarrow \mathbb{G}_T$ is modeled as a random oracle.

B. Correctness and Security Analysis

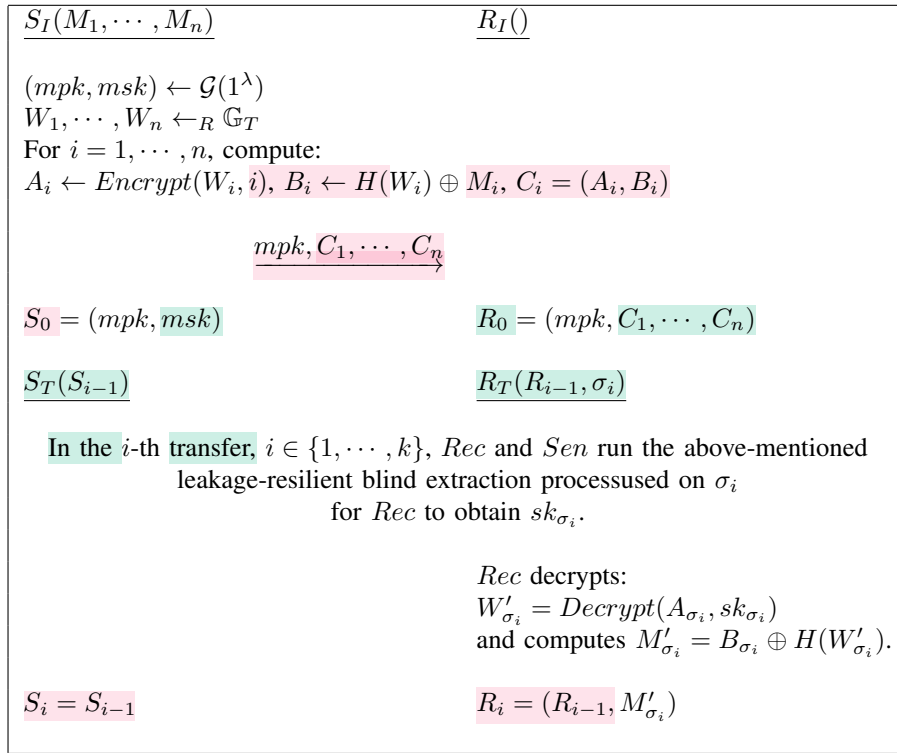
Correctness. The correctness of the adaptive $LR-OT_k^n$ protocol could be drawn from the correctness of $LR-OT_k^n$.

Security analysis. The following theorem establishes the security of the scheme:

Theorem 4.2. Assume that the assumption 1,2,3 and DL assumption hold, then the adaptive $LR-OT_k^n$ is one-side simulatable $(\log N - O(1), (0.15l - 3) \log N - 1)$ -leakage-resilient OT.

Proof sketch. We also separately prove security in the case where Rec is corrupted and the case where Sen is corrupted.

In the case of Rec is corrupted by a malicious adversary $\mathcal{A}$. We can construct an ideal-world simulator $\mathcal{S}_{Rec}$ as follows. $\mathcal{S}_{Rec}$ calls $Setup$ algorithm to get a master secret key msk and a master public key mpk , and substitutes the ciphertext vector $\langle C_1, \dots, C_n \rangle$ with a new vector $\langle C'_1, \dots, C'_n \rangle$, such that: $\langle A_1, \dots, A_n \rangle$ are computed as usual and $\langle$

Fig. 3. Adaptive LR- OT_k^n

$B_1, \dots, B_n$ are set to be random bitstrings. In the ideal-world, S_{Rec} simulates the random oracle, so S_{Rec} can extract σ_i when Rec calls $H(\cdot)$ on a value W_{σ_i} . Then S_{Rec} submits σ_i to the trust party and receives back the corresponding message M_{σ_i} . After that, S_{Rec} programs the random oracle so that $H(W_{\sigma_i}) = B_{\sigma_i} \oplus M_{\sigma_i}$.

During the protocol, the sender's secret memory includes msk , messages $M_1, \dots, M_n$ and random values $W_1, \dots, W_n$. So, if the adversary $\mathcal{A}$ submits a leakage function $Leak(\cdot)$ at any time, S_{Rec} defines a new leakage function $Leak^{sm}(\cdot)$ so that $Leak^{sm}(\cdot)$ is incorporated with the same code of $Leak(\cdot)$ except that msk and $W_1, \dots, W_n$ are already hardwired in it. Then S_{Rec} submits $Leak^{sm}(\cdot)$ to the trust party and receives back the leakage. Next, S_{Rec} hands $\mathcal{A}$ the leakage from the trust party. As above construction, $\lambda_S = \log N - O(1)$.

According to the security of the cryptosystem in [19], we claim $Real_{Leak(\cdot)}^A(Sen, Rec) \approx_c Ideal_{Leak^{sm}(\cdot)}^{S_{Rec}}(Sen, Rec)$.

In the case of Sen is corrupted by a semi-honest adversary $\mathcal{A}$. Intuitively, the receiver's behaviour in our Adaptive LR- OT_k^n protocol is almost the same as in LR- OT_k^n protocol, so the proof of our protocol's receiver security is same as the previous construction. For the sake of brevity, we omit it in our paper.

VI. CONCLUSION

In this paper, we present the first non-adaptive and adaptive leakage-resilient OT_k^n . Our constructions are based on Green et. al's OT protocols [18] and Lewko et. al's leakage-resilient cryptosystem [19]. The presented protocols achieve the half-simulatable security, and the adaptive protocol is constructed

in the random oracle model. So the open problem may be to construct an efficient adaptive LR- OT_k^n without random oracle.

REFERENCES

- [1] M. Rabin, 'How to exchange secrets by oblivious transfer'. Technical Report TR-81, Aiken Computation Laboratory, Harvard University, 1981.
- [2] A. Shamir, 'Identity-Based Cryptosystems and Signature Schemes'. Advances in Cryptology - CRYPTO 1984, vol. 196, pp. 47-53, August 1984.
- [3] A. Yao, 'How to generate and exchange secrets'. Proceedings of the 27th Annual Symposium on Foundations of Computer Science (FOCS 1986), pp. 162-167, 1986.
- [4] M. Naor and B. Pinkas, 'Oblivious transfer with adaptive queries'. Advances in Cryptology - CRYPTO 1999, vol. 1666, pp. 573-590, 1999.
- [5] J. Botelho, A. Nascimento, and R. Ramos, 'Commitment and Oblivious Transfer in the Bounded Storage Model With Errors', IEEE Transactions on Information Theory, vol. 71, no. 1, pp. 112-127, Jan. 2025.
- [6] Y. Zhou, R. Xu, Z. Qiao, and B. Yang, 'Leakage-resilient attribute-based encryption scheme with CCA security', Theoretical Computer Science, vol. 1059, 115648, Jan. 2026.
- [7] S. Dziembowski and S. Faust, 'Leakage-resilient cryptography from the inner-product extractor', Advances in Cryptology - ASIACRYPT 2011, vol. 7073, pp. 702-721, 2011.
- [8] Y. Chen, S. Luo, and Z. Chen, 'A new leakage-resilient IBE scheme in the relative leakage model', IEEE Transactions on Computers, vol. 74, no. 3, pp. 890-901, Mar. 2025.
- [9] J. H. Seo, J. Kim, and S. H. Hong, 'Leakage-resilient group signatures with dynamic membership', IEEE Transactions on Dependable and Secure Computing, vol. 22, no. 2, pp. 987-1002, Mar. 2025.
- [10] A. Cohen, S. Coretti, and Y. T. Kalai, 'Leakage-resilient extractors against number-on-forehead protocols', Proceedings of the 57th Annual ACM Symposium on Theory of Computing (STOC 2025), pp. 1899-1906, June 2025.
- [11] N. Bitansky, R. Canetti, and S. Halevi, 'Leakage-Tolerant Interactive Protocols', Theory of Cryptography Conference (TCC 2012), Lecture Notes in Computer Science, vol. 7194, pp. 266-284, March 2012.

- [12] E. Boyle, S. Garg, A. Jain, Y. T. Kalai, and A. Sahai, 'Secure Computation Against Adaptive Auxiliary Information,' *Advances in Cryptology - CRYPTO 2013*, vol. 8043, pp. 316-334, Aug. 2013.
- [13] I. B. Damgård, C. Hazay, and A. Patra, 'Leakage Resilient Secure Two-Party Computation,' *IACR Cryptology ePrint Archive*, Paper 2011/256, 2011.
- [14] P. S. Roy and A. Adhikari, 'One-sided leakage-resilient privacy only two-message oblivious transfer,' *Journal of Information Security and Applications*, vol. 19, no. 4-5, pp. 295-300, Nov. 2014.
- [15] C. Peikert, V. Vaikuntanathan, and B. Waters, 'A Framework for Efficient and Composable Oblivious Transfer,' *Advances in Cryptology - CRYPTO 2008*, vol. 5157, pp. 554-571, August 2008.
- [16] S. Halevi and H. Lin, 'Public-key cryptosystems resilient to key leakage,' *Advances in Cryptology - CRYPTO 2011*, vol. 6841, pp. 629-646, August 2011.
- [17] S. Micali and L. Reyzin, 'Physically Observable Cryptography (Extended Abstract),' *Theory of Cryptography Conference (TCC 2004)*, Lecture Notes in Computer Science, vol. 2951, pp. 278C296, February 2004.
- [18] M. Green and S. Hohenberger, 'Blind Identity-Based Encryption and Simulatable Oblivious Transfer,' *Advances in Cryptology - ASIACRYPT 2007*, vol. 4833, pp. 265-282, December 2007.
- [19] A. Lewko, Y. Rouselakis, and B. Waters, 'Achieving Leakage Resilience through Dual System Encryption,' *Theory of Cryptography Conference (TCC 2011)*, Lecture Notes in Computer Science, vol. 6597, pp. 70-88, March 2011.
- [20] Yuen, T.H., Chow, S.S.M., Zhang, Y. and Yiu, S.M.: 'Identity-Based Encryption Resilient to Continual Auxiliary Leakage'. *Proc. EUROCRYPT*, 2012, Springer (LNCS 7237), pp. 117-134.
- [21] J. Camenisch, G. Neven, and A. Shelat, 'Simulatable Adaptive Oblivious Transfer,' *Advances in Cryptology - EUROCRYPT 2007*, vol. 4515, pp. 573-590, May 2007.
- [22] S. Garg, A. Jain, and A. Sahai, 'Leakage-Resilient Zero Knowledge,' *Advances in Cryptology - CRYPTO 2011*, vol. 6841, pp. 297-315, August 2011.